Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 02)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen ☐

B Oxygen ☐

C Argon ☐

D Carbon dioxide ☐

0 1 . 2 Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

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0 1 . 3 Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

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0 2 . 1 Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops ☐

B Steam washing ☐

C Inflatable booms ☐

D Polymerising agents ☐

0 2 . 2 Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

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0 2 . 3 Explain why deoxygenation is a more common problem in water than in air.

[2 marks]

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0 3 . 1 Explain how each of the following human activities changes the amount of carbon in the atmosphere.

[6 marks]

Ploughing farmland .....

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Draining marshland / peat bogs .....

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Toxic pollution of the sea .....

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0 3 . 2 Give **three** anaerobic environments created by human activities that release methane.

[3 marks]

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0 3 . 3 Explain how rising atmospheric CO_2 causes ocean acidification.

[1 mark]

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0 4 A student is comparing the plant diversity of a grazed and an ungrazed grassland, and estimating the population of grasshoppers in each. In the grazed field 60 grasshoppers were caught, marked and released; three days later 75 were caught, of which 15 were marked.

0 4 . 1 Explain **three** reasons for choosing quadrat positions using random numbers rather than by eye.

[3 marks]

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0 4 . 2 Use the Lincoln index to estimate the grasshopper population of the grazed field.

[2 marks]

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Do not write outside the box

Answer

0 4 . 3 State **three** assumptions that must hold for this estimate to be valid.

[3 marks]

0 4 . 4 Explain how the student should decide the size of quadrat and the number of quadrats to use.

[3 marks]

0 4 . 5 Explain **four** other features of the design that would make the comparison between the two fields more reliable or more valid.

[4 marks]

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15

0 5 Table 1 shows measurements taken from a stream above and below the point where drainage water from an old metal-mine spoil heap enters it.

Table 1

Site	pH	Dissolved zinc / mg dm^{-3}	Dissolved oxygen / % saturation	Number of invertebrate taxa	Mayfly larvae present?
Upstream	7.1	0.02	96	18	Yes
50 m downstream	3.4	6.5	94	3	No
2 km downstream	5.9	1.1	95	9	No

0 5 . 1 Explain how the acidic drainage water is produced in the spoil heap.

[3 marks]

0 5 . 2 Use Table 1 to explain why the number of invertebrate taxa falls so sharply 50 m downstream, even though dissolved oxygen is unchanged.

[3 marks]

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0 5 . 3 Suggest why the pH and zinc concentration have partly recovered 2 km downstream.

[2 marks]

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0 5 . 4 Describe how acid mine drainage can be controlled.

[2 marks]

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0 6 . 1 Distinguish between **renewable** and **depletable** energy resources, giving an example of a resource that is both renewable and depletable.

[3 marks]

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0 6 . 2 Explain why the abundance of an energy resource is not the same as the amount available for use. Give **two** examples.

[3 marks]

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0 6 . 3 Describe the locational constraints of **two** named renewable energy resources.

[4 marks]

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0 7 A pesticide was applied to a field of oilseed rape. Table 1 shows the concentration of the pesticide in organisms sampled from a pond next to the field.

Table 1

Organism	Trophic level	Pesticide concentration / mg kg^{-1}
Pond water	-	0.002
Phytoplankton	1	0.04
Water fleas (Daphnia)	2	0.5
Small fish	3	4.2
Heron	4	38.0

0 7 . 1 Calculate how many times more concentrated the pesticide is in the heron than in the pond water. Give your answer to two significant figures.

[2 marks]

Answer ____ times

0 7 . 2 Name the process shown by the data in Table 1.

[1 mark]

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0 7 . 3 Explain the **two** properties of the pesticide that cause the pattern shown in Table 1.

[4 marks]

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0 7 . 4 The pesticide was probably an organochlorine. Suggest why the use of organochlorines is now banned or heavily restricted in most countries, even though DDT is still used to control malaria mosquitoes in some houses.

[3 marks]

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0 8 Table 1 shows the concentration of cadmium in rice and in the kidneys of people living in two villages in Japan. Village A is downstream of a zinc mine.

Table 1

Village	**Cadmium in rice / mg kg^{-1}**	**Cadmium in kidney of residents / mg kg^{-1}**	**Cases of bone disease (itai-itai) per 1000 people**
A (downstream of mine)	1.2	180	34
B (control)	0.05	22	0

0 8 . 1 Calculate how many times more concentrated cadmium is in the kidneys of village A residents than in their rice.

[1 mark]

Answer ..... times

0 8 . 2 Explain why cadmium and zinc pollution often occur together.

[1 mark]

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0 8 . 3 Use Table 1 and your knowledge of the properties of cadmium to explain the high rate of bone disease in village A.

[4 marks]

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0 8 . 4 Suggest **two** ways cadmium pollution from the mine could be controlled.

[2 marks]

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0 8 . 5 Cadmium is used in cadmium-telluride photovoltaic panels. Suggest why this new use may cause pollution in the future.

[2 marks]

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0 9 . 1 Explain how occupancy sensors and programmable thermostats reduce energy waste.

[2 marks]

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0 9 . 2 Explain why LED lights use much less electricity than the filament and CFL lamps they replaced, and give **two** other examples of low-energy appliance design.

[3 marks]

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0 9 . 3 Describe **three** changes in human behaviour that reduce domestic energy use.

[3 marks]

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0 9 . 4 Explain why heating water on demand saves energy compared with storing hot water.

[2 marks]

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1 0 . 1 Discuss how an understanding of the properties of pollutants can be used to predict their behaviour in the environment and to plan strategies to control them.

[9 marks]

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1 0 . 2 State what is meant by **bioaccumulation**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

OR

1 1 . 2 Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS